

Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

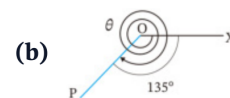
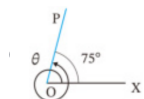
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Classified Practice

Defining Angles and Basic Trigonometric Functions

Q002 **Written****Draw the terminal side for 300° and -450° .**

Q003 Written

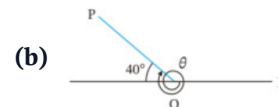
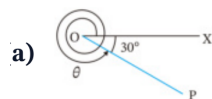
Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q005 Written

Draw the terminal side for 500° and -120° .

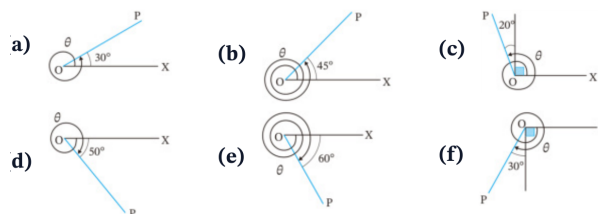
Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the source figure (a)–(f).

2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Q009 Written**Express** $610^\circ, 920^\circ, 840^\circ, -430^\circ, -960^\circ, -20^\circ$ **as** $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q010 Written

Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.

Q011 MCQ**The principal angle coterminal with 765° is:****Choose the correct option.**A 45° B 135° C 225° D 315°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Choose the correct option.**A -50° B 50° C 310° D 410°

Q013 MCQ**The terminal side of 540° lies on the:****Choose the correct option.**

- A positive x -axis
 - B negative x -axis
 - C positive y -axis
 - D negative y -axis
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Choose the correct option.**

A QI

B QII

C QIII

D QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Choose the correct option.**

- A $75^\circ + 360^\circ n$
 - B $285^\circ + 360^\circ n$
 - C $-285^\circ + 360^\circ n$
 - D $360^\circ n - 75^\circ$
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Choose the correct option.**

- A counter-clockwise
- B clockwise
- C radial only
- D undefined

Q017 MCQ**The terminal side of 1080° is the:****Choose the correct option.**

- A positive x -axis
- B negative x -axis
- C positive y -axis
- D negative y -axis

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Choose the correct option.**A $360^\circ n$ B $0^\circ n$ C $180^\circ + 360^\circ n$ D $90^\circ + 360^\circ n$

Q019 MCQ**The principal angle for -990° is:****Choose the correct option.**A 90° B 180° C 270° D -270°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Choose the correct option.

A 25°

B 155°

C 205°

D 335°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ **to degrees.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Q028 MCQ

Convert 150° to radians:**Choose the correct option.**

- A $\frac{5\pi}{6}$
B $\frac{3\pi}{5}$
C $\frac{5\pi}{3}$
D $\frac{3\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:**Choose the correct option.**A -120° B -150° C 150° D -300°

Q030 MCQ

For $r = 5$, $\theta = \frac{2\pi}{3}$, the arc length is:

Choose the correct option.

- A $\frac{5\pi}{3}$
- B $\frac{10\pi}{3}$
- C $\frac{25\pi}{3}$
- D 10π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Choose the correct option.

- A π
- B 2π
- C 4π
- D 8π

Q032 MCQ

The radian measure of a full turn is:**Choose the correct option.**

- A π
 - B 2π
 - C 180
 - D 360
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Choose the correct option.

A 210°

B 270°

C 315°

D 360°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Choose the correct option.

- A $\frac{\pi}{6}$
- B $-\frac{\pi}{6}$
- C $\frac{5\pi}{6}$
- D $\frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:**Choose the correct option.**

A $\frac{3\pi}{2}$

B 3π

C 9π

D $\frac{9\pi}{2}$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Choose the correct option.

A QI

B QII

C QIII

D QIV

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q041 **Written**

Find the exact values of the eighteen trigonometric functions listed in the source exercise (a)–(r), including negative and undefined cases.

Q042 Written

The panel height is $h(t) = 3 \sin\left(\frac{2\pi}{3}t - \frac{\pi}{4}\right) + 1.5$. Find the exact height at $t = 0$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q043 MCQ

 $\sin \frac{\pi}{6}$ equals:

Choose the correct option.

A 0

B $\frac{1}{2}$ C $\frac{\sqrt{3}}{2}$

D 1

Q044 MCQ $\cos \frac{2\pi}{3}$ equals:

Choose the correct option.

A -1

B $-\frac{1}{2}$ C $\frac{1}{2}$ D $\frac{\sqrt{3}}{2}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q045 MCQ

 $\tan \frac{3\pi}{4}$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Q046 MCQ $\sin\left(-\frac{\pi}{2}\right)$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q047 MCQ

 $\tan \frac{\pi}{2}$ is:

Choose the correct option.

A 0

B 1

C -1

D undefined

Q048 MCQ $\cos \frac{5\pi}{3}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:

Choose the correct option.

A x

B y

C x/y

D $1/x$

Q050 MCQ $\tan\left(-\frac{3\pi}{4}\right)$ equals:**Choose the correct option.**

A -1

B 0

C 1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q051 MCQ

 $\sin \frac{11\pi}{6}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Q052 Written

Find $\sin \frac{5\pi}{6}$, $\cos \frac{4\pi}{3}$, $\tan \frac{7\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q053 Written

Find $\sin(-\frac{5\pi}{6})$, $\cos(-\frac{7\pi}{6})$, $\tan(-\frac{\pi}{3})$.

Q054 Written

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Q056 Written**State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the source Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q059 **Written**

Solve the sixteen equations in the source Exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Q060 Written

Solve $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Choose the correct option.

- A $\frac{\pi}{6}, \frac{5\pi}{6}$
B $\frac{\pi}{3}, \frac{2\pi}{3}$
C $\frac{\pi}{6}, \frac{7\pi}{6}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{5\pi}{3}$
B $\frac{2\pi}{3}, \frac{4\pi}{3}$
C $\frac{\pi}{6}, \frac{5\pi}{6}$
D $\frac{4\pi}{3}, \frac{5\pi}{3}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Choose the correct option.**

- A $\frac{\pi}{4}, \frac{5\pi}{4}$
- B $\frac{3\pi}{4}, \frac{7\pi}{4}$
- C $\frac{\pi}{4}, \frac{7\pi}{4}$
- D $\frac{3\pi}{4}, \frac{5\pi}{4}$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Choose the correct option.**A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $0, 2\pi$ D π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:

Choose the correct option.

A 0

B $\frac{\pi}{2}$ C π D $\frac{3\pi}{2}$

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{4\pi}{3}$
B $\frac{2\pi}{3}, \frac{5\pi}{3}$
C $\frac{\pi}{6}, \frac{7\pi}{6}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:**Choose the correct option.**

- A $\frac{\pi}{2}$
- B $\frac{3\pi}{2}$
- C $\frac{2}{\pi}$
- D 0

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:

Choose the correct option.

- A $0, \pi$
- B $\frac{\pi}{2}, \frac{3\pi}{2}$
- C π
- D $\frac{\pi}{4}, \frac{5\pi}{4}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:**Choose the correct option.**A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $\frac{\pi}{4}, \frac{5\pi}{4}$ D $\pi, 2\pi$

Q070 Written**Solve** $\cos \theta = \frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Q072 Written**Solve** $\tan \theta = \frac{1}{\sqrt{3}}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Q074 Written**Solve** $2 \sin \theta + \sqrt{2} = 0$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)**The radian measure of 240° is:****Choose the correct option.**

- A $\frac{2\pi}{3}$
B $\frac{4\pi}{3}$
C $\frac{3\pi}{4}$
D $\frac{5\pi}{3}$
-
-
-

Quiz MCQ 2 Concept Quiz · MCQ

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{2\pi}{3}$
B $\frac{4\pi}{3}, \frac{5\pi}{3}$
C $\frac{3\pi}{2}, \frac{4\pi}{3}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 [Concept Quiz](#) · MCQ

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Choose the correct option.

A 3π

B 6π

C 12π

D 18π

Quiz MCQ 4 Concept Quiz · MCQ

The general-angle family for -300° is:

Choose the correct option.

- A $60^\circ + 360^\circ n$
- B $300^\circ + 360^\circ n$
- C $-60^\circ + 360^\circ n$
- D $120^\circ + 360^\circ n$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\cos \theta = -\frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Answer key

Use the question number shown on each page.

Q001 (a) 435° (b) -855°

Q002 300° : QIV

-450° : *negative y - axis*

Q003 $60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Q004 (a) -30° (b) 140°

Q005 500° : QI -120° : QIII

Q006 $160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Q007 30° ; 405° ; 110° ; 310° ; 300° ; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Q010 $120^\circ + 360^\circ n$; $260^\circ + 360^\circ n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Q022 $\frac{8\pi}{5}$, $-\frac{20\pi}{9}$; 330° , -108° ; $L = \frac{9\pi}{2}$, $S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}$, $\frac{\pi}{5}$, $-\frac{4\pi}{3}$, $-\frac{5\pi}{4}$, $\frac{7\pi}{3}$, $\frac{20\pi}{9}$, $-\frac{19\pi}{6}$, -15π

Q024 210° , 144° , 120° , 240° , -225° , -90° , -990° , -70°

Q025 (a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}$ cm; $V = 36\pi\sqrt{5}$ cm³

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi$, $S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $\sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $\frac{1}{\sqrt{3}}$; $-\frac{\sqrt{3}}{2}$; -1 ; *undefined*

Q041 $\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; 0 ; $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $-\frac{\sqrt{3}}{2}$; $-\sqrt{3}$; 1 ; 0 ; $-\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q042 $\frac{3(1-\sqrt{2})}{2}$ m

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}$, $-\frac{1}{2}$, -1

Q053 $-\frac{1}{2}$, $-\frac{\sqrt{3}}{2}$, $-\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $-\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}$, $\frac{3\pi}{2}$

Q057 (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$; (b) $\frac{\pi}{4}$, $\frac{7\pi}{4}$; (c) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

Q058 $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{7\pi}{6}$; 0 , π ; $\frac{\pi}{3}$, $\frac{5\pi}{3}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$

Q059 $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; 0 , π ; $\frac{4\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; $\frac{5\pi}{4}$, $\frac{3\pi}{4}$; $\frac{7\pi}{4}$, $\frac{5\pi}{4}$; $\frac{2\pi}{3}$, $\frac{4\pi}{3}$; $\frac{5\pi}{6}$, $\frac{11\pi}{6}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{4\pi}{3}$, $\frac{2\pi}{3}$; $\frac{3\pi}{2}$, $\frac{15\pi}{2}$

Q060 60° , 135° , 240° , 315°

Q061 $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{2}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}$, $\frac{5\pi}{3}$

Q072 $\frac{\pi}{6}$, $\frac{7\pi}{6}$

Q073 $\frac{\pi}{3}$, $\frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}$, $\frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}$, $\frac{5\pi}{4}$

Quiz WRITTEN 6 -390° ; 330°

Quiz WRITTEN 7 $L = 4\pi$, $S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q003 **Written**

For each of the twelve numbered cases in the source Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q005 Written

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q007 **Written**

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ in QIV, find $\sin \theta$ and $\cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q009 Written

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ in QIII, find $\cos \theta$ and $\tan \theta$.

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ in QIII, $\cos \theta$ is:

Choose the correct option.

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Choose the correct option.

- A $\frac{7}{24}$
- B $\frac{24}{7}$
- C $-\frac{7}{24}$
- D $\frac{25}{7}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ and θ in QIII, evaluate the exact value of $\sin^6 \theta - \cos^6 \theta$.

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in QII, find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ in QI, find $\sin \theta \cos \theta$ and $\sin^3 \theta - \cos^3 \theta$.

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Q023 Written**Given** $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{5}{16}$
B $\frac{11}{16}$
C $-\frac{11}{16}$
D $\frac{3}{16}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Choose the correct option.

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Choose the correct option.

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

Q029 Written

Prove $\frac{\sin \theta}{1 + \cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; (b) $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the source Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q034 Written

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

Q035 Written

Prove $\frac{\tan \theta}{1 + \tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}.$

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:**Choose the correct option.**

- A Change both sides at once
- B Simplify one side
- C Substitute random values
- D Take a square root

Q039 MCQ**The identity used to replace $1 - \sin^2 \theta$ is:****Choose the correct option.****A** $\cos^2 \theta$ **B** $\tan^2 \theta$ **C** $\sec^2 \theta$ **D** $\sin^2 \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q040 MCQ

$\tan \theta$ should be rewritten as:

Choose the correct option.

A $\sin \theta \cos \theta$

B $\sin \theta / \cos \theta$

C $\cos \theta / \sin \theta$

D $1 / \sin \theta$

Q041 MCQ**To add two trig fractions, first find a:****Choose the correct option.**

- A common numerator
- B common denominator
- C new quadrant
- D reference angle

Classified Practice

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Choose the correct option.**

- A LHS=0
- B LHS=RHS
- C RHS=1
- D undefined

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:

Choose the correct option.

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $-\frac{2\sqrt{5}}{5}$

C $\frac{\sqrt{5}}{5}$

D $-\frac{\sqrt{5}}{5}$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q047 MCQ**If $s + c = 1$, then sc equals:****Choose the correct option.****A** 0**B** $\frac{1}{2}$ **C** $-\frac{1}{2}$ **D** 1

Classified Practice

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{19}{27}$
B $\frac{5}{27}$
C $-\frac{19}{27}$
D $\frac{11}{27}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Choose the correct option.

- A $\frac{11}{16}$
B $-\frac{11}{16}$
C $\frac{7}{16}$
D $-\frac{7}{16}$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Choose the correct option.

- A $\frac{\sin \theta}{\cos \theta}$
- B $\frac{\cos \theta}{\sin \theta}$
- C $\sin \theta \cos \theta$
- D $\sec \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Choose the correct option.**A $\sin^2 \theta$ B $\tan^2 \theta$ C $\cos^2 \theta$

D 1

Q053 MCQ**When proving an identity, which side should normally be transformed?****Choose the correct option.**

- A Both sides simultaneously
- B One side only
- C Neither side
- D Only the denominator

Classified Practice

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:

Choose the correct option.

- A A random angle
- B The Pythagorean identity
- C A logarithm
- D A derivative

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin \theta + \cos \theta = \frac{1}{2}$, then $\sin \theta \cos \theta$ is:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Choose the correct option.

- A $\frac{3}{5}$
- B $\frac{4}{5}$
- C $-\frac{4}{5}$
- D $\frac{3}{4}$

Quiz MCQ 3 Concept Quiz · MCQ**Which identity is always valid where defined?****Choose the correct option.**

A $\tan \theta = \frac{\cos \theta}{\sin \theta}$

B $1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$

C $\sin \theta + \cos \theta = 1$

D $\sin \theta \cos \theta = 1$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written**Given** $s + c = \frac{2}{3}$, **find** sc **and** $s^3 + c^3$.

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}.$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Answer key

Use the question number shown on each page.

Q001 (a) $\cos \theta = -\frac{2\sqrt{6}}{5}$, $\tan \theta = \frac{\sqrt{6}}{12}$; (b) $\cos \theta = \frac{\sqrt{7}}{7}$, $\sin \theta = -\frac{\sqrt{42}}{7}$ **Q015** (a) $-\frac{3}{8}$ (b) $\frac{11}{16}$ (c) $\frac{\sqrt{7}}{2}$

Q002 (a) $\sin \theta = -\frac{\sqrt{6}}{3}$, $\tan \theta = \sqrt{2}$; (b) $\cos \theta = \frac{\sqrt{5}}{5}$, $\sin \theta = -\frac{2\sqrt{5}}{5}$; (c) $\cos \theta = -\frac{4}{5}$, $\tan \theta = -\frac{3}{4}$; (d) $\sin \theta = -\frac{\sqrt{21}}{7}$, $\cos \theta = -\frac{2\sqrt{7}}{7}$

Q003 $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (b) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (c) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (d) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$ **Q016** (a) $-\frac{4}{9}$ (b) $\frac{37}{27}$ (c) $\frac{\sqrt{17}}{3}$

Q004 $\sin \theta = -\frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$ or the swapped pair **Q017** (1) $\sin \theta \cos \theta = \frac{1}{8}$, $\sin^2 \theta - \cos^2 \theta = \frac{7\sqrt{3}}{16}$, $\sin \theta + \cos \theta = \frac{\sqrt{5}}{2}$; (2) $-\frac{1}{8}$, $\frac{9\sqrt{3}}{16}$, $-\frac{\sqrt{5}}{2}$; (3) $\frac{3}{8}$, $\frac{11}{16}$ **Q031** LHS = RHS for (a)+(f)

Q005 $-\frac{12}{13}$, $-\frac{5}{12}$ **Q018** $-\frac{55\sqrt{7}}{256}$ **Q032** LHS = $2 \csc x$

Q006 $-\frac{4}{5}$, $\frac{4}{3}$ **Q019** $-\frac{5}{18}$, $\frac{\sqrt{14}}{3}$ **Q033** Identity proved

Q007 $\frac{3}{5}$, $\frac{4}{5}$ **Q020** $-\frac{1}{8}$, $\frac{7\sqrt{5}}{16}$ **Q034** Identity proved

Q008 $-\frac{5}{13}$, $\frac{12}{13}$ **Q021** $-\frac{11}{16}$ **Q035** Identity proved

Q009 $-\frac{\sqrt{2}}{2}$, 1 **Q022** $\frac{\sqrt{11}}{3}$ **Q036** Identity proved

Q010 B **Q023** $\frac{3}{2}$ **Q037** Identity proved

Q011 A **Q024** A **Q038** B

Q012 B **Q025** B **Q039** A

Q013 C **Q026** B **Q040** B

Q014 A **Q027** A **Q041** B

Q028 A

Q029 LHS = RHS

Q030 Both identities verified

Q031 LHS = RHS for (a)+(f)

Q032 LHS = $2 \csc x$

Q033 Identity proved

Q034 Identity proved

Q035 Identity proved

Q036 Identity proved

Q037 Identity proved

Q038 B

Q039 A

Q040 B

Q041 B

Q042 B

Q043 A

Q044 A

Q045 B

Q046 D

Q047 A

Q048 A

Q049 A

Q050 A

Q051 B

Q052 A

Q053 B

Q054 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $-\frac{12}{13}$, $\frac{12}{5}$

Quiz WRITTEN 6 $-\frac{5}{18}$, $\frac{19}{27}$

Quiz WRITTEN 7 Identity proved

Quiz WRITTEN 8 $\sin \theta = -\frac{2\sqrt{5}}{5}$, $\cos \theta = \frac{\sqrt{5}}{5}$; $6 = 6$

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three source demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q004 **Written**

Simplify $\sin\left(\frac{3\pi}{2} - \theta\right) \sec(\pi + \theta) - \tan\left(\theta - \frac{\pi}{2}\right) \cot\left(\frac{5\pi}{2} + \theta\right).$

Q005 Written

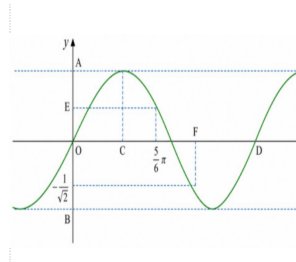
Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, and $y = \tan \theta$, and state each period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q006 Written

For the source graph of $y = \sin \theta$, determine the labelled scales A to F .

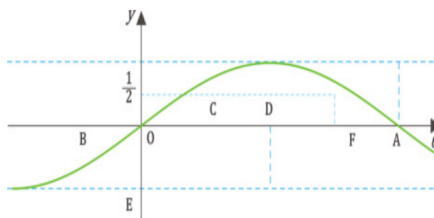


Classified Practice

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the source graph of $y = \cos \theta$, determine the labelled scales A to F .

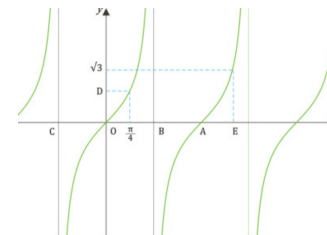
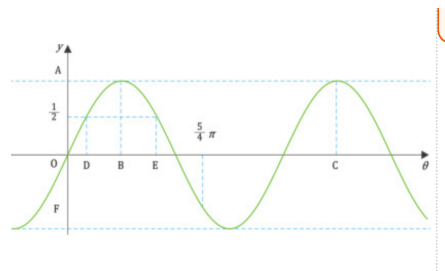
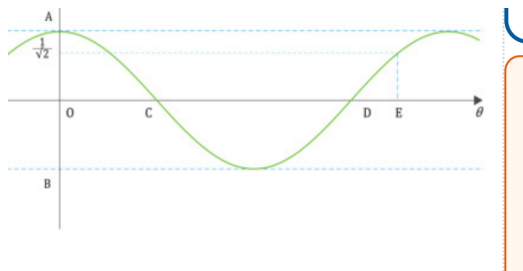


Classified Practice

Properties and Graphs of Trigonometric Functions

Q008 **Written**

Complete all four source graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the source cosine graph; read the labelled scales on the source sine graph; and read the labelled scales on the source tangent graph.

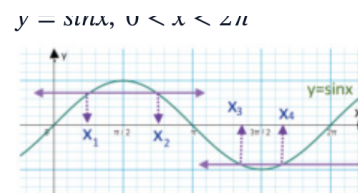


Classified Practice

Properties and Graphs of Trigonometric Functions

Q009 Written

The source graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine source graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.

Q014 Written**Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q015 Written

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all source graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos \left| x + \frac{\pi}{3} \right|$ and find its period.

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine source graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.

Q022 MCQ $\sin(\theta + \pi)$ equals:**Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q024 MCQ $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:

Choose the correct option.

A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q026 MCQ **$\sin(-\theta)$ equals:****Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q028 MCQ $\sin\left(\frac{3\pi}{2} + \theta\right)$ **equals:****Choose the correct option.**A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:

Choose the correct option.

A $\tan \theta$ B $-\tan \theta$ C $\cot \theta$ D $-\cot \theta$

Q030 MCQ**The period of $y = \sin \theta$ is:****Choose the correct option.****A** $\pi/2$ **B** π **C** 2π **D** 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Choose the correct option.**

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D $y > 1$

Q032 MCQ

The period of $y = \tan \theta$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Choose the correct option.**

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Choose the correct option.**

A -1

B 0

C 1

D π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Choose the correct option.**

- A $\pi/4$
- B $\pi/2$
- C π
- D $3\pi/2$

Q036 MCQ**At $\theta = 0$, $\cos \theta$ equals:****Choose the correct option.****A** -1**B** 0**C** 1**D** undefined

Classified Practice

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Choose the correct option.**

- A -1
- B 0
- C 1
- D undefined

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Choose the correct option.**A $[-1,1]$ B $[-3,3]$ C $[0,3]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Choose the correct option.**A $\pi/4$ B $\pi/2$ C π D 2π

Q040 MCQ**The period of $y = \cos(\theta/3)$ is:****Choose the correct option.**A $2\pi/3$ B 2π C 3π D 6π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Choose the correct option.**A $\pi/5$ B π C 5π D $2\pi/5$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:

Choose the correct option.

A $[-2, 2]$

B $[-1, 1]$

C $[-1/2, 1/2]$

D $[0, 1/2]$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Choose the correct option.**

- A shifted only
- B reflected and stretched vertically
- C compressed horizontally only
- D undefined

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C 3π D 6π

Q046 MCQ**The graph of $y = \sin(\theta - \pi/4)$ is shifted:****Choose the correct option.**

- A left by $\pi/4$
- B right by $\pi/4$
- C up by $\pi/4$
- D down by $\pi/4$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Choose the correct option.**A right by $\pi/3$ B left by $\pi/3$ C up by $\pi/3$ D down by $\pi/3$

Q048 MCQ

A horizontal shift changes the sine period from 2π to:**Choose the correct option.**A π B 2π C 4π

D 0

Classified Practice

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Q050 MCQ**The graph of $y = \sin(\theta + \pi/2)$ equals:****Choose the correct option.**A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Choose the correct option.**A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:**Choose the correct option.**

- A $\theta = 0$
- B $\theta = \pi/2$
- C $\theta = \pi$
- D $\theta = 2\pi$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Choose the correct option.**A $\cos(x)$ B $-\cos(x)$ C $\sin(x)$ D $|\cos(x)|$

Q054 MCQ**In $y = a \sin(b\theta - q)$, the horizontal shift is:****Choose the correct option.**

- A** q
- B** q/b
- C** b/q
- D** a/b

Classified Practice

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C π D 2π

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Choose the correct option.**A $[-1, 1]$ B $[-2, 2]$ C $[-4, 4]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Choose the correct option.**

- A left by $\pi/3$
- B right by $\pi/6$
- C right by $\pi/3$
- D left by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Choose the correct option.**

- A π
- B 2π
- C 3π
- D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:

Choose the correct option.

- A vertical scale
- B horizontal shift
- C horizontal scale
- D reflection only

Q060 MCQ**The period of $y = \tan(4\theta + q)$ is:****Choose the correct option.**A $\pi/4$ B $\pi/2$ C π D 2π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Choose the correct option.**A $[-3,3]$ B $[0,3]$ C $[-1,1]$ D $\mathbb{R}$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos\left(\frac{\pi}{2} - \theta\right)$.

Q064 Written**Simplify** $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan\left(-\frac{3\pi}{4}\right)$ **using an acute reference angle.**

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Q070 Written**Find the period and range of $y = 4 \cos \theta$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q071 Written

Find the period and range of $y = \sin(5\theta)$.

Q072 Written**Find the period and range of $y = \tan(\theta/4)$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Q074 Written**Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

Q076 Written**Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Q078 Written**For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 Concept Quiz · MCQ

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Quiz MCQ 2 Concept Quiz · MCQ

$\sin(\theta + \pi)$ equals:

Choose the correct option.

A $\sin \theta$

B $-\sin \theta$

C $\cos \theta$

D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)**The range of $y = 2 \cos \theta$ is:****Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $\mathbb{R}$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Choose the correct option.

- A left by $\frac{\pi}{4}$
- B right by $\frac{\pi}{4}$
- C up by $\frac{\pi}{4}$
- D down by $\frac{\pi}{4}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written**Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.**

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\max} = 2\pi, T_{\min} = 2\pi, T_{\text{mid}} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q001 Written

Solve (a) $-2 \cos \theta = \sqrt{2}$ and (b) $\tan \theta + \sqrt{3} = 0$, giving the general solution for (b).

Q002 Written

Solve (a) $2 \cos \theta - 1 = 0$ and (b) $\tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all source equations on $0 \leq \theta \leq 2\pi$: (a) $2 \sin \theta + 1 = 0$, (b) $2 \sin \theta = \sqrt{3}$, (c) $3 \sin \theta + 3 = 0$, (d) $\sqrt{2} \cos \theta + 1 = 0$, (e) $2 \cos \theta = \sqrt{3}$, (f) $2 \cos \theta - \sqrt{2} = 0$, (g) $\tan \theta + 1 = 0$, (h) $\sqrt{3} \tan \theta = 1$, (i) $\sqrt{3} \tan \theta = 3$, and give general solutions for (j) $2 \sin \theta + \sqrt{3} = 0$, (k) $\cos \theta + 1 = 0$, (l) $\sqrt{3} \tan \theta + 3 = 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q004 **Written**

Solve $\tan^2 \theta - (2 + \sqrt{3}) \tan \theta + 2\sqrt{3} = 0$ and give the general solution.

Q005 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos \theta > -\sqrt{3}$, (b) $\tan \theta \leq \frac{1}{\sqrt{3}}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q006 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin \theta \leq -1$, (b) $\cos \theta > \frac{1}{2}$, (c) $\tan \theta > 1$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q007 Written

Solve all source inequalities for $0 \leq \theta < 2\pi$: (a) $\sin \theta > \frac{\sqrt{3}}{2}$, (b) $\sin \theta - \frac{\sqrt{2}}{2} > 0$, (c) $2 \sin \theta + \sqrt{3} \leq 0$, (d) $\cos \theta > 0$, (e) $\cos \theta - \frac{1}{2} \leq 0$, (f) $2 \cos \theta + \sqrt{2} \leq 0$, (g) $\tan \theta \leq -\frac{1}{\sqrt{3}}$, (h) $\tan \theta + \sqrt{3} > 0$, (i) $\tan \theta + 1 \geq 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q008 Written

Solve $2 \sin^2 \theta - \cos \theta - 1 > 0$ **for** $0 \leq \theta < 2\pi$.

Q009 Written**Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos^2 \theta - 3 \sin \theta = 0$, (b) $2 \sin^2 \theta - 3 \cos \theta - 3 \leq 0$.**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta - \sin \theta = 0$, (b) $2 \cos^2 \theta - \sin \theta = 2$, (c) $2 \sin^2 \theta - 4 < 5 \cos \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q011 Written

Solve all source equations and inequalities for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta + \sin \theta - 1 = 0$, (b) $2 \sin^2 \theta + 3 \cos \theta = 0$, (c) $2 \sin^2 \theta - \cos \theta = 2$, (d) $2 \cos^2 \theta - 5 \cos \theta - 3 = 0$, (e) $2 \cos^2 \theta + \sin \theta - 1 = 0$, (f) $2 \cos^2 \theta + 3 \sin \theta = 3$, (g) $2 \cos^2 \theta + 3 \sin \theta \leq 0$, (h) $2 \sin^2 \theta + \cos \theta > 1$, (i) $2 \cos^2 \theta - 4 \leq 5 \sin \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Q013 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, (b) $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q014 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q016 Written

Solve $4 \sin^2\left(\theta + \frac{\pi}{6}\right) \geq 1$ for $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2 \cos x}$. Evaluate this strategy and propose a better substitution on $[0^\circ, 360^\circ[$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q018 MCQ

The period used for a general tangent solution is:**Choose the correct option.**A 2π B π C $\pi/2$ D 4π

Q019 MCQ**The solutions of $2 \cos \theta = 1$ on $[0, 2\pi]$ are:****Choose the correct option.****A $\pi/6, 11\pi/6$** **B $\pi/3, 5\pi/3$** **C $2\pi/3, 4\pi/3$** **D $\pi/4, 7\pi/4$**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q020 MCQ

Which root is impossible when solving $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$?

Choose the correct option.

A $\sin \theta = 1/2$

B $\sin \theta = -2$

C $\sin \theta = 0$

D $\sin \theta = -1/2$

Q021 MCQ

The quadratic challenge factors into roots:**Choose the correct option.**

- A 1 and $\sqrt{3}$
- B 2 and $\sqrt{3}$
- C 2 and 3
- D $\pi/3$ and 2

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q022 MCQ

For $2 \cos \theta > -\sqrt{3}$, the excluded arc is:

Choose the correct option.

A $[0, 5\pi/6]$

B $[5\pi/6, 7\pi/6]$

C $[7\pi/6, 2\pi)$

D $(\pi/6, 5\pi/6)$

Q023 MCQ**The inequality $\tan \theta > 1$ on $[0, 2\pi)$ holds on:****Choose the correct option.****A** $(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$ **B** $[0, \pi/4]$ **C** $(\pi/2, \pi)$ **D** $[\pi, 2\pi)$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q024 MCQ

A closed endpoint in a trig inequality is used when the sign is:

Choose the correct option.

- A strictly greater
- B strictly less
- C greater than or equal to
- D never

Q025 MCQ**The 5-14 challenge reduces to which cosine condition?****Choose the correct option.**

A $\cos \theta > 1/2$

B $-1 < \cos \theta < 1/2$

C $\cos \theta \leq -1$

D $\cos \theta \geq 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q026 MCQ

When a quadratic gives $\sin \theta = -2$, we:**Choose the correct option.**

A keep it

B reject it

C square it

D divide by 2

Q027 MCQ

The first step for $2 \cos^2 \theta - 3 \sin \theta = 0$ is to use:**Choose the correct option.**

A $\sin^2 \theta + \cos^2 \theta = 1$

B $\tan \theta = \sin \theta$

C $\cos \theta = 1$

D the tangent period

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q028 MCQ

The solution of $2 \sin^2 \theta + \sin \theta - 1 = 0$ includes:

Choose the correct option.

A $\theta = \pi/6$

B $\theta = \pi/4$

C $\theta = \pi/3$

D $\theta = 2\pi/3$

Q029 MCQ**In the rational challenge, which angle must be excluded?****Choose the correct option.****A** 0**B** $\pi/4$ **C** $\pi/2$ **D** π

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Choose the correct option.

A $[0, 2\pi)$

B $[\pi/3, 7\pi/3)$

C $[-\pi/3, 5\pi/3)$

D $[\pi, 3\pi)$

Q031 MCQ

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

Choose the correct option.

A $\pi/12$

B $\pi/6$

C $\pi/3$

D $\pi/2$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ -solution is:

Choose the correct option.

A $[0, \pi/2]$

B $(0, \pi/2)$

C $[\pi/2, \pi]$

D $[0, 2\pi)$

Q033 MCQ**The 5-16 challenge is equivalent to:****Choose the correct option.**

A $|\sin(t)| \geq 1/2$

B $\sin(t) = 0$

C $\cos(t) > 1/2$

D $\tan(t) = 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ simplifies to:**Choose the correct option.**A $\cos(x)$ B $\sin(x)$ C $\tan^2 x$

D 1

Quiz MCQ 1 **Concept Quiz · MCQ**

The general solution of $\tan \theta = -1$ is:

Choose the correct option.

A $-\pi/4 + 2n\pi$

B $-\pi/4 + n\pi$

C $\pi/4 + 2n\pi$

D $\pi/2 + n\pi$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 2 **Concept Quiz · MCQ**

The solution of $\cos \theta > 0$ on $[0, 2\pi)$ is:

Choose the correct option.

A $(\pi/2, 3\pi/2)$

B $[0, \pi/2) \cup (3\pi/2, 2\pi)$

C $[0, \pi)$

D $(\pi, 2\pi)$

Quiz MCQ 3 Concept Quiz · MCQ

The valid sine root of $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ is:

Choose the correct option.

A $\sin \theta = -2$

B $\sin \theta = 1/2$

C $\sin \theta = 2$

D $\sin \theta = -1/2$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The first operation in a shifted-argument problem is to:

Choose the correct option.

- A divide by $\cos \theta$
- B set t equal to the whole argument and find its range
- C change radians to degrees
- D square both sides

Quiz WRITTEN 5 Concept Quiz · Written

Solve $2 \cos \theta = \sqrt{3}$ for $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\tan \theta \leq -1$ for $0 \leq \theta < 2\pi$.

Quiz WRITTEN 7 Concept Quiz · Written

Solve $2 \sin^2 \theta + \sin \theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

Q001 (a) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ (b) $\theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$ **Q012** $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q020** B

Q002 (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ **Q013** (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q021** B

Q003 (a) $\frac{3\pi}{4}, \frac{11\pi}{4}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q014** (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q022** B

Q004 $\tan \theta = 2$ or $\tan \theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$ **Q015** (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q023** A

Q005 (a) $0 \leq \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ (b) $0 \leq \theta \leq \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \leq \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$ **Q016** $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q024** C

Q006 (a) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (b) $0 \leq \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ (c) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$ **Q017** Dividing by $\cos(x)$ separates the

Q007 (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q018** B

Q008 $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$ **Q019** B

Q009 (a) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ (b) $0 \leq \theta \leq \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \leq \theta < 2\pi$ **Q020** B

Q010 (a) $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ (b) $\theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $0 \leq \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ **Q021** B

Q011 (a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (e) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (g) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (h) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (i) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (j) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (k) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (l) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (m) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (n) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (o) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (p) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (q) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (r) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (s) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (t) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (u) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (v) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (w) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (x) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (y) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (z) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ **Q022** B

Q012 $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q023** A

Q013 (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q024** C

Q014 (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q025** B

Q015 (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q026** B

Q016 $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q027** A

Q017 Dividing by $\cos(x)$ separates the interacting factors and is not helpful, **Q028** A

Q018 B **Q029** C

Q019 B **Q030** B

Q020 B **Q031** A

Q021 B **Q032** A

Q022 B **Q033** A

Q023 A **Q034** B

Q024 C **Quiz MCQ 1** B

Q025 B **Quiz MCQ 2** B

Q026 B **Quiz MCQ 3** B

Q027 A **Quiz MCQ 4** B

Q028 A **Quiz WRITTEN 5** $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Q029 C **Quiz WRITTEN 6** $\frac{\pi}{2} < \theta \leq \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$

Q030 B **Quiz WRITTEN 7** $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Q031 A **Quiz WRITTEN 8** $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Classified Practice

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Q002 Written

Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Classified Practice

Advanced Theorems and Applications

Q003 Written

Solve the four source maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the source challenge $y = \cos^2 \theta - \sin \theta$.

Classified Practice

Advanced Theorems and Applications

Q004 **Written****Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.**

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ **in QII** and $\cos \beta = -\frac{3}{5}$ **in QIII**, **find** $\sin(\alpha + \beta)$ **and** $\cos(\alpha - \beta)$.

Classified Practice

Advanced Theorems and Applications

Q006 Written

Complete the source addition-theorem exercises *exact values, quadrant – controlled* α, β , and *the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Q007 Written**Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.**

Classified Practice

Advanced Theorems and Applications

Q008 Written

Find the angle between each source pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Classified Practice

Advanced Theorems and Applications

Q009 Written

Find the exact angles for all source line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Classified Practice

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ with α in QII, find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Classified Practice

Advanced Theorems and Applications

Q012 **Written**

Solve the source double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Q013 Written

Given $\cos \alpha = 3/5$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\cos(5\pi/8)$.

Classified Practice

Advanced Theorems and Applications

Q014 **Written**

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Q015 Written

Solve the source half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi,$
 $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi.$

Classified Practice

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q017 Written**Solve the source equation** $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q018 Written

Solve all source double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Classified Practice

Advanced Theorems and Applications

Q020 Written

Rewrite the source expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in harmonic-addition form.

Q021 Written

Rewrite all source harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Classified Practice

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.

Q023 Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q024 **Written**

Solve all source harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Q025 Written

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the source functions $y = -\frac{4}{3} \sin^2 \theta + \cos \theta$ and $y = \cos 2\theta - 2 \cos \theta$.

Q027 Written

Solve all source maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Classified Practice

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:

Choose the correct option.

A $t = \sin \theta$

B $t = \cos \theta$

C $t = \tan \theta$

D $t = \theta$

Q029 MCQ**The maximum of $\sin^2 \theta + \cos \theta + 1$ is:****Choose the correct option.****A** 0**B** 1**C** 9/4**D** 4

Classified Practice

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Choose the correct option.**A $[-2,2]$ B $[-1,1]$ C $[0,1]$

D all real numbers

Q031 MCQ

The identity for $\sin(a+b)$ is:**Choose the correct option.**

- A $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
- B $\sin \alpha \sin \beta$
- C $\cos \alpha \cos \beta$
- D $\tan \alpha + \tan \beta$

Classified Practice

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Choose the correct option.**A $(\sqrt{6}-\sqrt{2})/4$ B $(\sqrt{6}+\sqrt{2})/4$ C $1/2$ D $\sqrt{3}/2$

Q033 MCQ**A missing trig value is selected using:****Choose the correct option.**

- A the quadrant
- B the denominator
- C the period only
- D a graph scale

Classified Practice

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Choose the correct option.**

A k

B m

C $1/m$ D $m+k$

Q035 MCQ**The angle formula uses denominator:****Choose the correct option.****A** $1+m_1m_2$ **B** m_1-m_2 **C** m_1+m_2 **D** m_1m_2

Classified Practice

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Choose the correct option.

A θ_1

B $\pi - \theta_1$

C $2\theta_1$

D $-\theta_1$ only

Q037 MCQ **$\sin(2a)$ equals:****Choose the correct option.**

- A $\sin \alpha$
- B $2 \sin \alpha \cos \alpha$
- C $\cos^2 \alpha$
- D $\tan \alpha$

Classified Practice

Advanced Theorems and Applications

Q038 MCQ

 $\cos(2\alpha)$ can be written as:**Choose the correct option.**A $\cos^2 \alpha - \sin^2 \alpha$ B $2\sin \alpha$ C $\tan \alpha$ D $1 + \sin \alpha$

Q039 MCQ**For theta in QIV, sin theta is:****Choose the correct option.**

- A positive
- B negative
- C zero always
- D undefined

Classified Practice

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Choose the correct option.**

- A $(1 - \cos \alpha)/2$
- B $(1 + \cos \alpha)/2$
- C $\tan \alpha$
- D $\cos 2\alpha$

Q041 MCQ**The sign of a half-angle root is chosen from:****Choose the correct option.**

- A the original quadrant interval
- B the coefficient only
- C the denominator
- D the period of $\tan$

Classified Practice

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Choose the correct option.

A negative

B positive

C zero

D undefined

Q043 MCQ**For double-angle equations, first rewrite in:****Choose the correct option.**

- A one trig function
- B two unrelated angles
- C degrees only
- D a logarithm

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Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Choose the correct option.**A $\sin\theta(2\cos\theta + 1)$ B $\cos\theta(2\sin\theta - 1)$ C $\sin\theta + \cos\theta$ D $2\sin\theta$

Q045 MCQ**A tangent asymptote is always:****Choose the correct option.**

- A included
- B excluded
- C a maximum
- D an endpoint

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Q046 MCQ

For $a \sin \theta + b \cos \theta$, r is:**Choose the correct option.**A $a+b$ B $\sqrt{a^2+b^2}$ C $a-b$ D ab

Q047 MCQ**The harmonic target form is:****Choose the correct option.****A** $r \sin(\theta + \alpha)$ **B** $r \cos \theta$ **C** $a \tan \theta$ **D** $\sin \theta$ only

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Q048 MCQ

The angle a is located from point:**Choose the correct option.**

- A $P(a,b)$
- B $P(r,\theta)$
- C the origin only
- D the x intercept

Q049 MCQ**After synthesis, t is usually:****Choose the correct option.****A** $\theta + \alpha$ **B** θ^2 **C** $\tan \theta$ **D** $\cos \theta$

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Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < \frac{3}{2}$ is:

Choose the correct option.

- A never true
- B always true
- C true only at π
- D undefined

Q051 MCQ**In a strict inequality, equality endpoints are:****Choose the correct option.**

- A included
- B excluded
- C doubled
- D random

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Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $[-3,3]$

Q053 MCQ**For a double-angle maximum problem, use:****Choose the correct option.**

- A the double-angle formula then a bounded variable
- B a logarithm
- C only a graph
- D no substitution

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Q054 MCQ

A vertex outside $[-1,1]$ is:**Choose the correct option.**

A an allowed extremum

B rejected

C always the maximum

D always the minimum

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

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Quiz MCQ 2 Concept Quiz · MCQ

The addition-theorem value $\sin 75^\circ$ is:

Choose the correct option.

A $(\sqrt{6} + \sqrt{2})/4$

B $(\sqrt{6} - \sqrt{2})/4$

C $1/2$

D 1

Quiz MCQ 3 Concept Quiz · MCQ

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Choose the correct option.

A -2

B -1

C 1

D 2

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Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Choose the correct option.

A $\pi/6$

B $\pi/4$

C $\pi/3$

D $\pi/2$

Quiz WRITTEN 5 Concept Quiz · Written

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

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Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and** α **is in** QII.

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

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Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{2}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a\theta = \frac{\pi}{2}, \pi$; $b\ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}, \min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$